Income Tax Calculator Coverage Report

Lab 6 | Code and Branch Coverage

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Result

The income tax module reached 100% line coverage and 100% branch coverage. The line suite passed 21 tests; the branch suite passed 24 tests and exercised all 28 decision branches with no partial branches.

Coverage summary

Suite	Tests passed	Statements	Branches	Coverage
Code / line	21	39 / 39	Not measured	100%
Branch	24	39 / 39	28 / 28	100%

Coverage was measured for income_tax.py only. The tests cover valid and invalid inputs, both tax regimes, the investment cap, age exemption boundaries, tax brackets, and every refund or balance outcome.

Reproduce the results

Run these commands from the LAB6 folder:

```
python -m coverage erase
python -m coverage run --source=income_tax -m pytest CodeCoverageTest_income_tax_application.py -q
python -m coverage report -m
python -m coverage erase
python -m coverage run --branch --source=income_tax -m pytest BranchCoverageTest_income_tax_application.py -q
python -m coverage report -m
```

Commits

fd86ae9 Add income tax line coverage tests
1da21de Add income tax branch coverage tests

Code coverage evidence

```
CODE COVERAGE RUN OUTPUT

.....................                                                [100%]
============================== warnings summary ===============================
..\..\..\..\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\sarth\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14:
PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
21 passed, 1 warning in 0.32s
Name           Stmts   Miss  Cover   Missing
----------------------------------------------
income_tax.py      39      0   100%
----------------------------------------------
TOTAL              39      0   100%
```

Coverage output transcript image: 21 passing tests and 100% line coverage.

Branch coverage evidence

```
BRANCH COVERAGE RUN OUTPUT

.....................                                                [100%]
============================== warnings summary ===============================
..\..\..\..\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\sarth\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14:
PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
24 passed, 1 warning in 0.18s
Name           Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------
income_tax.py      39      0     28      0   100%
-----------------------------------------------------------
TOTAL              39      0     28      0   100%
```

Coverage output transcript image: 24 passing tests, 28 branches, and 100% branch coverage.